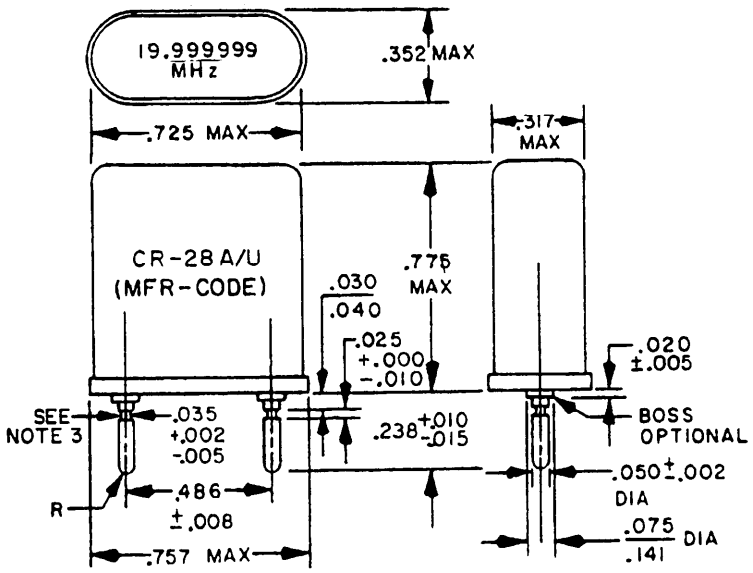
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-28A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; controlled; series.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The pin undercut may be omitted.

(E) denotes changes.

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure
Frequency range	0.8 to 20 MHz, inclusive
Capacitance, shunt	7 pF maximum
Frequency tolerance:	
Operating temperature range	±20 PPM
Room temperature	±70 PPM
Frequency stability	± 5 PPM
Equivalent resistance	See table II
Mode of oscillation	Fundamental
Reference temperature	75° ±1°C
(E) Temperature ranges:	
Operable	-55° to +70° and +80° to +90° C inclusive
Operating	70° to 80° C. inclusive
Test set, calibration values, and rated drive level	See table III.
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
(E) Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent-resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 201 of MIL-STD-202.

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TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	520
0.85+ to 0.9	480
0.9+ to 1	440
1+ to 1.12	400
1.12+ to 1.25	380
1.25+ to 1.37	340
1.37+ to 1.5	300
1.5+ to 1.62	270
1.62+ to 1.75	250
1.75+ to 1.87	220
1.87+ to 2	180
2+ to 2.12	160
2.12+ to 2.25	150
2.25+ to 2.6	120
2.6+ to 3	90
3+ to 3.4	70
3.4+ to 3.75	52
3.75+ to 4	45
4+ to 5	37
5+ to 7	25
7+ to 10	20
10+ to 15	18
15+ to 20	15

TABLE III. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	ohms	mA	volt	mW
TS-330/TSM	0.8 to 1.5	0	1/	---	5.0 ±1.0 2/
	1.5+ to 2.25	50	10	---	5.0 ±1.0
	2.25+ to 3.4	50	10	---	5.0 ±1.0
	3.4+ to 5.1	22	15	---	5.0 ±1.0
	5.1+ to 7.5	13	20	---	5.0 ±1.0
	7.5+ to 10	13	20	---	5.0 ±1.0
	10+ to 15	11	15	---	2.5 ±0.5
TS-683/TSM	15+ to 20	10	---	0.16	2.5 ±0.5

1/ Set crystal current control at extreme counterclockwise (minimum) position.
2/ The violation of $P = I^2R$ is intentional; at the higher resistance of the crystal unit, the empirical power dissipation will be as rated.

NOTES:
1. The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/4 (Sockets, Crystal, Top and Bottom Mounting, 2 Contacts). For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO preferred type NXT8/A.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU, SM, SG
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0511-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098/10E CRYSTAL UNITS, QUARTZ, CR-28A/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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